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# Syndromic Surveillance Of Infectious Diseases

## ***Priority Pathogens To Be Tested For In Indian Clinical Settings***

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DIVISION OF COMMUNICABLE DISEASES

INDIAN COUNCIL OF MEDICAL RESEARCH, NEW DELHI, INDIA

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### Priority Pathogens To Be Tested For In Indian Clinical Settings

#### A. **Background:**

The laboratory testing algorithms for infectious diseases currently followed in India are varied, institute/region-specific, and largely directed towards individual diseases with sequential testing, with limited availability of scientific literature on countrywide prevalence and trends of pathogens to guide India-specific surveillance and testing policies. Consequently, pathogen testing in clinical microbiology laboratories differs as per institutional requirements and available laboratory resources, often missing out on common/critical pathogens, thereby hampering patient care and aggravating public health problems such as antimicrobial resistance. To address this issue, the ICMR has listed important pathogens causing critical infectious syndromes in India, in an effort to guide patient management, public health policies, and biomedical R&D.

#### B. **Methodology:**

The priority pathogen lists have been developed in consultation with eminent clinicians, laboratory experts, and epidemiologists from across the country over a series of 44 meetings. Working groups of subject experts have conducted extensive review of literature from India and other similar settings, and drawn on their individual experiences to draft syndrome-wise lists of priority pathogens based on their prevalence, trends, outbreak potential, disease elimination targets, and availability of diagnostics/therapeutics. These lists have been reviewed, edited and approved by eminent review committee members.

#### C. **Applications and Benefits:**

This document lists priority pathogens causing ***Acute Febrile Illness (including Acute Undifferentiated Fever, Fever with Rash, Fever with Lymphadenopathy), Acute Encephalitis Syndrome, Acute Respiratory Illness, and Acute Diarrhoeal Disease***, and is intended to guide medical practitioners, laboratory experts, epidemiologists and policymakers on the diagnostic and surveillance approaches for these syndromes. These lists are also meant to guide indigenous industry partners for development of customized India-specific cost-effective medical countermeasures of public health importance.

The pathogens are listed in a graded approach, from commonly encountered microbes to rare/exotic/high-risk pathogens, which is described below:

1. **Priority 1A:** Commonly detected pathogens in clinical settings that are crucial to diagnose before initiating treatment.
2. **Priority 1B:** Other commonly detected pathogens in clinical settings
3. **Priority 2:** Less common pathogens found in clinical settings
4. **Priority 3:** Rare pathogens, which may not be detected in routine clinical settings and may be detected during specialized laboratory testing as a part of surveillance

## 1. PRIORITY PATHOGENS CAUSING ACUTE FEBRILE ILLNESS

### i. Acute Undifferentiated Febrile Illness (AUFI):

**Case definition:** Patient with fever  $\geq 38.0^{\circ}\text{C}$  or  $100.4^{\circ}\text{F}$  of  $\leq 14$  days duration without localization (i.e., without evidence of localized infection by history, physical examination, investigations such as complete blood count, biochemistry profile, urine analysis, or chest radiography) at the time of initial presentation.

| Priority 1A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Priority 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Priority 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| <ol style="list-style-type: none"> <li>1. <i>Plasmodium sp.</i></li> <li>2. Dengue virus<sup>^@</sup></li> <li>3. <i>Salmonella Typhi</i></li> <li>4. <i>Salmonella Paratyphi A, B</i></li> <li>5. <i>Orientia tsutsugamushi</i></li> <li>6. Influenza virus (Influenza A (H1N1)pdm09, A(H3N2), Influenza B)<sup>^@</sup></li> <li>7. <i>Leptospira sp.</i><sup>*@^</sup></li> <li>8. Chikungunya virus<sup>^@</sup></li> <li>9. <i>Brucella melitensis</i><sup>#</sup></li> <li>10. Chandipura virus</li> </ol> | <ol style="list-style-type: none"> <li>1. <i>Borrelia burgdorferi</i></li> <li>2. Other Rickettsial infections (<i>R. typhi</i>, <i>R. prowazekii</i>, <i>R. conori</i>, <i>R. akari</i>, <i>R. rickettsii</i>)<sup>\$</sup></li> <li>3. <i>Listeria monocytogenes</i></li> <li>4. West Nile Virus<sup>@</sup></li> <li>5. <i>Parvovirus B19</i></li> <li>6. Zika Virus<sup>@</sup></li> <li>7. Crimean Congo Haemorrhagic Fever virus<sup>*@</sup></li> <li>8. Kyasanur Forest Disease virus<sup>*@</sup></li> <li>9. Epstein-Barr Virus</li> <li>10. Cytomegalovirus</li> <li>11. <i>Neisseria meningitidis</i><sup>@</sup></li> <li>12. <i>Bartonella henselae</i><sup>##</sup></li> <li>13. <i>Streptobacillus moniliformis</i><sup>###</sup></li> <li>14. <i>Spirillum minus</i><sup>###</sup></li> <li>15. <i>Trypanosoma sp.</i> (Zoonotic Trypanosomiasis)</li> <li>16. <i>Aspergillus sp.</i><sup>@@</sup></li> <li>17. <i>Candida sp.</i><sup>@@</sup></li> <li>18. Enteroviruses including Coxsackieviruses<sup>\$</sup></li> <li>19. Adenovirus<sup>\$</sup></li> <li>20. Human Herpesvirus-6<sup>\$</sup></li> </ol> | <ol style="list-style-type: none"> <li>1. <i>Bacillus anthracis</i><sup>*@</sup></li> <li>2. <i>Yersinia pestis</i><sup>@</sup></li> <li>3. <i>Francisella tularensis</i><sup>####</sup></li> <li>4. <i>Histoplasma capsulatum</i><sup>**</sup></li> <li>5. Hantavirus<sup>θ</sup></li> <li>6. Ebola virus<sup>θ</sup></li> <li>7. Marburg virus<sup>θ</sup></li> <li>8. Yellow Fever virus<sup>θ</sup></li> <li>9. Langya henipavirus<sup>θ</sup></li> <li>10. Rift Valley Fever<sup>θ</sup></li> <li>11. Lymphocytic Choriomeningitis Virus<sup>θ</sup></li> <li>12. <i>Coccidioides sp.</i></li> </ol> |
| Priority 1B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <ol style="list-style-type: none"> <li>1. Common bacteria causing sepsis (ESKAPE)</li> <li>2. <i>Burkholderia pseudomallei</i></li> <li>3. <i>Mycobacterium tuberculosis</i><sup>**</sup></li> <li>4. Atypical Mycobacteria (<i>M. avium complex</i>, <i>M. chelonae</i>, <i>M. fortuitum</i>, <i>M. abscessus</i>, etc)<sup>**</sup></li> <li>5. <i>Leishmania donovani</i><sup>*</sup></li> <li>6. Filarial worm (Lymphatic Filariasis)<sup>*</sup></li> </ol>                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

\*High priority in endemic areas; @ Outbreak prone – high priority during outbreak; ^ Seasonal – high priority during the season (post monsoon; flooding etc); \$ a subset of test negative samples to be screened in order to understand the prevalence; # High priority if history of exposure to animals- ## Cat, ### Rat #### Rabbit; θ Exotic; \*\*Not a common cause of AUFI and to suspect only when lab investigations are not specific and radiological signs are observed; @@ Suspect only if the patient is prone to infections (e.g. immunocompromised; cancer, transplant patients etc.)

## Syndromic Surveillance of Infectious Diseases

### Priority Pathogens To Be Tested For In Indian Clinical Settings

#### ii. Fever with Rash:

| Priority 1                                                                                                                                                                                                                                                                                                                                                                                | Priority 2                                                                                                                                                                                                                                              | Priority 3                                                                                                                                                                                                                                                                                                                                                        |
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| <ol style="list-style-type: none"> <li>Measles virus @</li> <li>Rubella virus</li> <li><i>Orientia tsutsugamushi</i> &amp; Other Rickettsia</li> <li>Dengue Virus^@</li> <li>Varicella Zoster Virus</li> <li>Herpes Simplex Virus</li> <li>Enterovirus (Hand Foot &amp; Mouth Disease)^@</li> <li>Zika Virus^@</li> <li>Parvovirus B19</li> <li><i>Neisseria meningitidis</i>@</li> </ol> | <ol style="list-style-type: none"> <li>Rickettsial Pox</li> <li>Human Herpesvirus-6</li> <li>Epstein-Barr Virus</li> <li><i>Borrelia burgdorferi</i> (Lyme Disease)</li> <li>West Nile virus</li> <li>Chandipura Virus</li> <li>Mpox Virus +</li> </ol> | <ol style="list-style-type: none"> <li><i>Spirillum minus</i>, <i>Streptobacillus moniliformis</i> ###</li> <li><i>Francisella tularensis</i> ####</li> <li>SARS-CoV-2 <sup>Ω</sup></li> <li>Hanta Virus</li> <li>Ebola &amp; Marburg viruses</li> <li>Yellow Fever virus</li> <li>Rift Valley Fever virus</li> <li>Lymphocytic Choriomeningitis Virus</li> </ol> |
| <b>1B</b>                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                   |
| <ol style="list-style-type: none"> <li>Chikungunya virus^@</li> <li><i>Leptospira sp.</i> ^@*</li> <li>Salmonella Typhi/ Paratyphi</li> <li>Common Bacterial causes- (Staphylococcal Scalded Skin Syndrome, Scarlet fever etc)</li> <li>Crimean Congo Hemorrhagic Fever virus*@</li> <li>Kyasanur Forest Disease virus*@</li> </ol>                                                       | <ol style="list-style-type: none"> <li>&amp; Other pox viruses#####</li> <li>Cytomegalovirus</li> <li><i>Bartonella sp.</i> ##</li> <li><i>Bacillus anthracis</i> #✓</li> </ol>                                                                         |                                                                                                                                                                                                                                                                                                                                                                   |

\*High priority in endemic areas; @ Outbreak prone – high priority during outbreak; ^ Seasonal – high priority during the particular season (post monsoon; flooding etc); # High priority if history of exposure to animals- ## Cat (and skin pustule at the site of bite/scratch) , ### Rat (Rat bite fever), #### Rabbit, ##### Cow, Buffalo, Sheep, Goat etc; + High priority if history of contact with a case/sexual contact/promiscuous behaviour ✓ Skin lesion/ulcer/eschar; Ω MISC of COVID-19

**iii. Fever with Lymphadenopathy:**

| Priority 1A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Priority 2                                                                                                                                                                                                                                                                                                                                    | Priority 3                                                                                                                                                                                                                                                                                                                                                                                |
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| <ol style="list-style-type: none"> <li>1. Epstein-Barr virus (Infectious Mononucleosis like syndrome) *</li> <li>2. Cytomegalovirus (Infectious Mononucleosis like syndrome) *</li> <li>3. <i>Orientia tsutsugamushi</i></li> <li>4. <i>Brucella melitensis</i>#</li> <li>5. <i>Burkholderia pseudomallei</i>^</li> <li>6. Other <i>Rickettsial</i> sp. – e.g.: <i>Rickettsia akari</i> causing Rickettsial pox</li> <li>7. <i>Mycobacterium tuberculosis</i></li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                  | <ol style="list-style-type: none"> <li>1. <i>Bartonella henselae</i>##</li> <li>2. <i>Toxoplasma gondii</i>##</li> <li>3. <i>Coxiella burnetii</i> (Q fever)#</li> <li>4. <i>Yersinia pestis</i>@</li> <li>5. <i>Francisella tularensis</i>####</li> <li>6. Hepatitis B Virus</li> <li>7. Hepatitis C Virus</li> <li>8. Mpox Virus</li> </ol> | <ol style="list-style-type: none"> <li>1. <i>Histoplasma capsulatum</i></li> <li>2. <i>Coccidioides</i> sp.</li> <li>3. <i>Paracoccidioides</i> sp.</li> <li>4. <i>Blastomyces dermatitidis</i></li> <li>5. <i>Cryptococcus</i> sp.</li> <li>6. Other conditions commonly associated with generalized lymphadenopathy but not acute fever</li> <li>7. <i>Vaccinia virus</i> \$</li> </ol> |
| Priority 1B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                           |
| <ol style="list-style-type: none"> <li>1. Rubella virus*</li> <li>2. Measles virus *</li> <li>3. Adenovirus*</li> <li>4. Common bacteria causing pyogenic infections (<i>S. aureus</i>/ <i>Streptococcus pyogenes</i>/ <i>Anaerobic bacteria</i>, etc)</li> <li>5. <i>Corynebacterium diphtheriae</i>*</li> <li>6. Filarial worm (Lymphatic filariasis) <math>\Omega</math></li> <li>7. <i>Leishmania donovani</i> <math>\Omega</math></li> <li>8. Non-tuberculous <i>Mycobacteria</i> (<i>M. avium</i> complex etc.)</li> <li>9. Human Herpesvirus 6</li> <li>10. Herpes Simplex Virus <math>\alpha\beta</math></li> <li>11. <i>Treponema pallidum</i> <math>\beta</math></li> <li>12. Human Immunodeficiency Virus</li> <li>13. Typhoidal <i>Salmonellae</i>-<i>S. Typhi/Paratyphi</i></li> <li>14. Varicella Zoster Virus</li> <li>15. Human Herpes Virus 8</li> </ol> |                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                           |

\*More common in pediatric population; # High priority if history of exposure to animals- ## Cat, #### Rabbit; @ Outbreak prone – High Priority during outbreak; ^ exposure to the coastal soil and working in paddy fields; \$ immunocompromised;  $\Omega$  - High priority in endemic areas;  $\alpha$  – associated with oropharyngeal lesions;  $\beta$  – associated with genital lesions

## 2. PRIORITY PATHOGENS CAUSING ACUTE ENCEPHALITIS SYNDROME (AES)

### Case Definition

A case of AES is defined by the National Center for Vector Borne Diseases Control (NCVBDC) as a person of any age, presenting with the below symptoms at any time of the year:

Acute onset of fever ( $\leq 7$  days)

AND

Change in mental status (including symptoms such as confusion, disorientation, coma or inability to talk)

AND /OR

New onset of seizures (excluding simple febrile seizures)

| Priority 1A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Priority 1B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Priority 2                                                                                                                                                                       |
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| <p><u>Most Common Pathogens/Ease of availability of diagnostic tests/ Inclusion in National Programme</u></p> <ol style="list-style-type: none"> <li>Japanese Encephalitis virus</li> <li><i>Plasmodium</i> sp.</li> <li><i>Orientia tsutsugamushi</i> &amp; other <i>Rickettsial</i> sp.</li> </ol> <p><u>If Viral aetiology suspected (based on CSF &amp; Radiology Findings )</u></p> <ol style="list-style-type: none"> <li>Herpes Simplex virus I &amp; II</li> <li>Dengue virus@^</li> <li>Chikungunya virus@^</li> <li>Enteroviruses@</li> <li>West Nile virus* @</li> <li>Epstein-Barr virus</li> <li>Varicella Zoster virus</li> <li>Human Herpesvirus 6</li> <li>Parvovirus B19</li> <li>Adenovirus</li> <li>Chandipura virus* @</li> <li>Nipah virus* @</li> <li>Measles virus (SSPE) @</li> <li>Kyasanur Forest Disease virus@</li> <li>Crimean Congo Haemorrhagic Fever virus @</li> <li>Rabies virus #</li> <li>Mumps virus</li> <li>Cytomegalovirus</li> <li>Rubella virus</li> <li>Human Immunodeficiency virus</li> </ol> | <p><u>If Bacterial aetiology is suspected (based on CSF &amp; Radiology Findings )</u></p> <ol style="list-style-type: none"> <li><i>Streptococcus pneumoniae</i></li> <li>Group B Streptococcus (<i>Streptococcus agalactiae</i>)</li> <li><i>Neisseria meningitidis</i></li> <li><i>Haemophilus influenzae</i></li> <li>Gram Negative bacteria (<i>E.coli</i>, <i>Klebsiella</i> sp., <i>Pseudomonas</i> sp., etc)</li> <li><i>Leptospira</i> sp.</li> <li><i>Brucella</i> sp. #</li> <li><i>Borrelia burgdorferi</i></li> <li><i>Listeria monocytogenes</i></li> <li><i>Mycobacterium</i> sp.</li> <li><i>Treponema pallidum</i> (Syphilis)</li> </ol> <p><u>Parasitic</u></p> <ol style="list-style-type: none"> <li><i>Taenia</i> sp. (Cysticercosis)</li> <li><i>Toxoplasma gondii</i></li> <li><i>Naegleria</i> sp./ <i>Acanthamoeba</i> sp.~</li> </ol> <p><u>Fungal</u></p> <ol style="list-style-type: none"> <li><i>Cryptococcus</i> sp.</li> <li><i>Aspergillus</i> sp.</li> <li><i>Candida</i> sp.</li> </ol> | <ol style="list-style-type: none"> <li>Human Herpesvirus 7</li> <li>Zika Virus</li> <li>Lymphocytic Choriomeningitis virus θ</li> <li>Tick borne encephalitis virus θ</li> </ol> |

\*High Priority in endemic areas; @ Outbreak prone – High Priority during outbreak; ^ Seasonal – High Priority during the season (Post monsoon; flooding etc); # High priority if history of exposure to Animals; θ Exotic; ~ H/o Swimming in ponds & other water bodies, canals

### **3. PRIORITY PATHOGENS CAUSING ACUTE RESPIRATORY ILLNESS (ARI)**

#### **Case Definition**

Influenza Like Illness (ILI) and Severe Acute Respiratory Infection (SARI) case definitions may be employed for surveillance studies for detection of influenza, SARS-CoV-2 and other viral aetiologies.

ILI is defined by the World Health Organization (WHO) as an acute respiratory infection with:

- measured fever of  $\geq 38\text{ C}^\circ$
- and cough;
- with onset within the last 10 days

SARI is defined by WHO as an acute respiratory infection with:

- history of fever or measured fever of  $\geq 38\text{ C}^\circ$
- and cough;
- with onset within the last 10 days,
- and requires hospitalization.

## Syndromic Surveillance of Infectious Diseases

### Priority Pathogens To Be Tested For In Indian Clinical Settings

| Priority 1A                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                              | Priority 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Priority 3                                                                                                                                                                                                               |
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| <b>Viruses</b> <ol style="list-style-type: none"> <li>Seasonal Influenza viruses - Influenza A &amp; Influenza B</li> <li>SARS-CoV-2</li> <li>Respiratory Syncytial Virus</li> <li>Adenovirus</li> <li>Human Respiroviruses 1 &amp; 3 and Human Rubulaviruses 2 &amp; 4 (Formerly Human Parainfluenzaviruses 1-4) @</li> <li>Human Metapneumovirus @</li> <li>Measles virus +</li> </ol>                                                                   | <b>Bacteria</b> <ol style="list-style-type: none"> <li><i>Streptococcus pneumoniae</i></li> <li><i>Mycoplasma pneumoniae</i></li> <li><i>Corynebacterium diphtheriae</i></li> <li><i>Haemophilus influenzae</i></li> <li><i>Mycobacterium tuberculosis</i>+</li> <li>Gram negative bacilli (<i>K. pneumoniae</i>, <i>P. aeruginosa</i> etc.)</li> <li><i>Bordetella pertussis</i></li> </ol> | <ol style="list-style-type: none"> <li>Anaerobic bacteria</li> <li>Non-tuberculous Mycobacteria</li> <li>Varicella Zoster virus</li> <li>Herpes simplex virus I &amp; II</li> <li><i>Moraxella catarrhalis</i></li> <li><i>Arcanobacterium haemolyticum</i></li> <li><i>Yersinia enterocolitica</i></li> <li><i>Legionella pneumophila</i></li> <li><i>Bacillus anthracis</i> **</li> <li><i>Chlamydia psittaci</i> #</li> <li><i>Coxiella burnetii</i></li> <li><i>Burkholderia pseudomallei</i></li> <li><i>Neisseria gonorrhoeae</i></li> <li>Hantavirus ###</li> <li>Avian influenza virus #</li> <li>Epstein-Barr virus</li> <li><i>Ascaris lumbricoides</i></li> <li><i>Talaromyces</i> sp.</li> <li><i>Histoplasma capsulatum</i></li> <li><i>Coccidioides immitis</i></li> <li><i>Blastomyces dermatitidis</i></li> <li><i>Candida</i> sp.</li> </ol> | <ol style="list-style-type: none"> <li><i>Francisella tularensis</i> ##</li> <li><i>Yersinia pestis</i> ###</li> <li>MERS</li> <li>Coronavirus</li> <li><i>Strongyloides stercoralis</i></li> <li>Nipah virus</li> </ol> |
| Priority 1B                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                          |
| <ol style="list-style-type: none"> <li>Rhinovirus</li> <li><i>Streptococcus pyogenes</i></li> <li><i>Staphylococcus aureus</i></li> <li><i>Chlamydia pneumoniae</i></li> <li>Group C and G Streptococci</li> <li><i>Corynebacterium ulcerans</i></li> <li>Human Bocavirus</li> <li>Enteroviruses</li> <li>Cytomegalovirus \$</li> <li><i>Cryptococcus neoformans</i> *</li> <li><i>Aspergillus</i> sp.*</li> <li><i>Pneumocystis jirovecii</i>*</li> </ol> |                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                          |

\* More common with immunocompromised hosts, \$ More common among infants born to CMV infected mother, transplant patients and immunocompromised patients,\*\* To be prioritized in endemic region or patients with exposure history (cattle, sheep, goat) @ More severe in children + In view of National elimination program; High priority if history of exposure to animals- # Birds, ## Rabbit, ### Rodents



#### **4. PRIORITY PATHOGENS CAUSING ACUTE DIARRHOEAL DISEASE**

##### **Case Definition**

Diarrhoea has been defined by WHO as passage of three or more loose or liquid stools per day (or more frequent passage than is normal for the individual).

| Priority 1A                                                                                                                                                                                                                                                                                                                                                                                                                         | Priority 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Priority 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ol style="list-style-type: none"> <li>1. Diarrheagenic <i>E. coli</i> #@</li> <li>2. Non-typhoidal <i>Salmonella</i> @</li> <li>3. <i>Shigella</i> sp. @</li> <li>4. <i>Vibrio cholerae</i></li> <li>5. Norovirus (genogroup I, II)</li> <li>6. Rotavirus</li> <li>7. <i>Giardia lamblia</i> #</li> <li>8. <i>Entamoeba histolytica</i> @</li> <li>9. Adenovirus (types 40, 41)</li> <li>10. <i>Campylobacter</i> sp. @</li> </ol> | <ol style="list-style-type: none"> <li>1. <i>Yersinia enterocolitica</i> @</li> <li>2. <i>Listeria</i> sp.</li> <li>3. <i>Aeromonas</i> sp. @</li> <li>4. <i>Plesiomonas</i> sp. @</li> <li>5. Non cholera <i>Vibrio</i> sp.</li> <li>6. Astrovirus</li> <li>7. Sapovirus</li> <li>8. Microsporidium #</li> <li>9. <i>Dientamoeba fragilis</i></li> <li>10. <i>Taenia</i> sp.</li> <li>11. <i>Strongyloides stercoralis</i> #</li> <li>12. <i>Cyclospora</i> sp. #</li> <li>13. <i>Cystoisospora belli</i> #</li> <li>14. Hookworm (<i>Ancylostoma</i> sp., <i>Necator americanus</i>)</li> <li>15. <i>Ascaris lumbricoides</i></li> <li>16. <i>Trichuris trichiura</i></li> </ol> | <ol style="list-style-type: none"> <li>1. <i>Staphylococcus aureus</i> \$</li> <li>2. <i>Bacillus cereus</i> \$</li> <li>3. Cytomegalovirus ^#</li> <li>4. Herpes simplex virus I &amp; II ^</li> <li>5. <i>Blastocystis hominis</i></li> <li>6. <i>Balantidium coli</i> @</li> <li>7. <i>Schistosoma</i> sp. (Intestinal Schistosomiasis) @</li> <li>8. <i>Fasciolopsis buski</i></li> <li>9. <i>Clonorchis sinensis</i></li> <li>10. <i>Heterophyes heterophyes</i></li> <li>11. <i>Metagonimus yokogawai</i></li> <li>12. <i>Paragonimus westermani</i></li> <li>13. <i>Gastrodiscoides hominis</i></li> <li>14. <i>Trichinella spiralis</i></li> <li>15. <i>Capillaria philippinensis</i></li> <li>16. <i>Angiostrongylus</i> sp.</li> </ol> |
| Priority 1B                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <ol style="list-style-type: none"> <li>1. <i>Cryptosporidium</i> sp. #</li> <li>2. <i>Hymenolepis nana</i></li> <li>3. <i>Clostridioides difficile</i> *@</li> </ol>                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

@ cause of dysentery; # Persistent diarrhea, in Immunocompromised; \* antibiotic associated diarrhea;

^ suspected in immunocompromised, tested with biopsy samples; \$ vomiting may be predominant symptom, due to preformed toxin

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